

Kaitlyn Hill
Ivan Wu

Final Report on Game Store Database

The original deployment was hosted on an Oregon State University class server. The hosting environment was temporary and is no longer active.

Index

Executive Summary.....	2
Project Outline.....	3
Database Outline.....	3
ERD.....	5
Schema.....	5
Sample Data.....	6
Screen Captures and CRUD operations.....	8
Home Page.....	8
Products Page.....	9
Customers Page.....	11
Employees Page.....	12
Sales Page.....	13
Sale Items Page.....	14

Executive Summary

This project focused on designing and implementing a relational database and web application for a small game store. The goal was to create a system that could manage product, employees, customers, and sale transactions while allowing users to perform CRUD (CREATE, READ, UPDATE, DELETE) operations through a web application. The project was completed in multiple stages, with each step allowing time for peer review and feedback. This process helped identify areas for improvement and guided changes to the database design, SQL structure, and web application interface.

During the early design stages, several changes were made. The original design included a separate Inventory table that tracked the quality and quantity of each product for a M:1 relationship. However, this table was quickly removed from the database to simplify the structure, while quantity and quality went directly into the product table. The schema moving forward contained five entities: product, sale, employee, customer, and product_sale. The sale table records each transaction, while the product_sale table lists the specific products sold in that transaction and how many were purchased. Additional feedback helped improve attribute choices and data consistency. Attribute fields such as platform, genre, quality, and employee role data types were changed to ENUM to keep data consistent. A rewards system was also added to give purpose to the customer table; it tracks rewards earned, rewards redeemed, and reward balance.

Later steps in the project focused on improving SQL data definition language (DDL) structure and implementing database constraints. Reviews and feedback helped identify small issues such as duplicate values in sample data, minor typos, schema formatting, and database normalization. Foreign key constraints were added to maintain relationships between tables. In particular, ON DELETE CASCADE was used so that when a sale is deleted, the related rows in the product_sale table are automatically removed. However, some delete restrictions were intentionally kept to preserve historical sales data. For example, this prevented the deletion of a customer if they were associated with a sale, the sale would have to be deleted first in order for the customer to be deleted.

The web application interface was then developed to display and manage database records. Each entity has its own page where users can view and manage data. At this point in the project, the majority of edits/updates focused on readability, aesthetics and layout (formatting, aligning headers, and adding grid lines to make the information easier to read). A RESET button was also added to restore the database to its original state for testing CRUD operations. In the final stage, the web application was completed so users could add, view, update, and delete records directly through the interface, with input validation to prevent incorrect data entry.

AI tools were used very minimally in this project. The database design, SQL implementation, and web application development were all original work. AI was used to help walkthrough a Node.js issue as some of us were having trouble directly accessing the database. Overall, AI served only as a limited support tool rather than for generating core code or design decisions.

Project Outline

A small store that sells video games and game related merchandise is struggling to keep track of their inventory, sales, etc. A database driven website will help consolidate their resources and help them monitor several attributes. The store has around 5,000 sale transactions annually, which the website will record the sale to customer, as well as the specific product and product_sale. Employees will be able to look up the quantity of each product as well as the quality of it for customers. During a transaction, the sale table records all the information the sale (who did it, who bought it (if applicable), and how much) while the product_sale table records the information of items in the sale (products quantity/quality/price).

Database Outline

product - Specific item sold (Game, board game, etc)

- product_ID: int, PK, not null, unique, auto-increment
- product_name: varchar(200), not null
- platform: enum(), not null
 - Playstation, Xbox, Nintendo, PC, Board Game, Card Game, Other
- genre: enum(), not null
 - Action, Adventure, RPG, Shooter, Sports, Simulation, Puzzle, Horror, Fighting, Racing, Sandbox, Survival, Strategy, Party, Other
- price: decimal(10,2), not null
- quality: enum(), not null
 - Unsellable, Poor, Average, Excellent, Near Mint
- quantity: int, not null, default 0
- Relationships
 - product to product_sale (1:M), implementation of product_ID as FK into product_sale.

customer - Tracks customer information

- customer_ID: int, PK, not null, unique, auto_increment
- first_name: varchar(50)
- last_name: varchar(50)
- email: varchar(100), unique
- phone: varchar(10), unique
 - Constraint that either email or phone number needs to be NOT NULL
- reward_balance: decimal(10,2), not null, default 0
 - Customers gain reward points at a 1:10 ratio, i.e. \$10 spent = \$1 reward points. Max \$500 in reward points
- Relationship
 - customer to sale (1:M), implementation of customer_ID as FK in sale

employee - Tracks employee information

- employee_ID: int, PK, not null, unique, auto_increment
- first_name: varchar(50), not null
- last_name: varchar(50), not null
- role: enum()
 - Cashier, Stocker, Manager, Supervisor, Other
- Relationship
 - Employee to Sale (1:M)

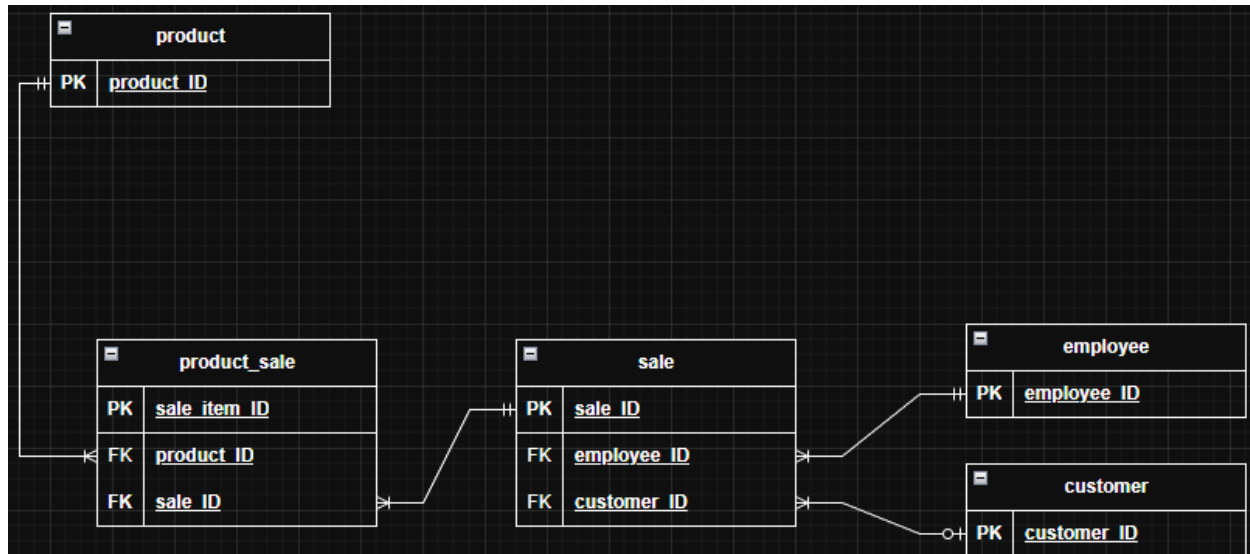
sale - Tracks one complete transaction (general information)

- sale_ID: int, PK, not null, unique, auto_increment
- sale_date: datetime, not null
- total_price: decimal(10,2), not null
- customer_ID: int, FK
- employee_ID: int, FK, not null
- rewards_earned: decimal(10,2)
 - Constraint, not null if customer_ID is not null
- rewards_redeemed: decimal(10,2)
 - Constraint, not null if customer_ID is not null
- Relationship
 - sale to product_sale (1:M), implementation of sale_ID as FK into product_sale
 - sale to employee (M:1), implementation of employee_ID as FK into sale
 - sale to customer (M:1), implementation of customer_ID as FK into sale

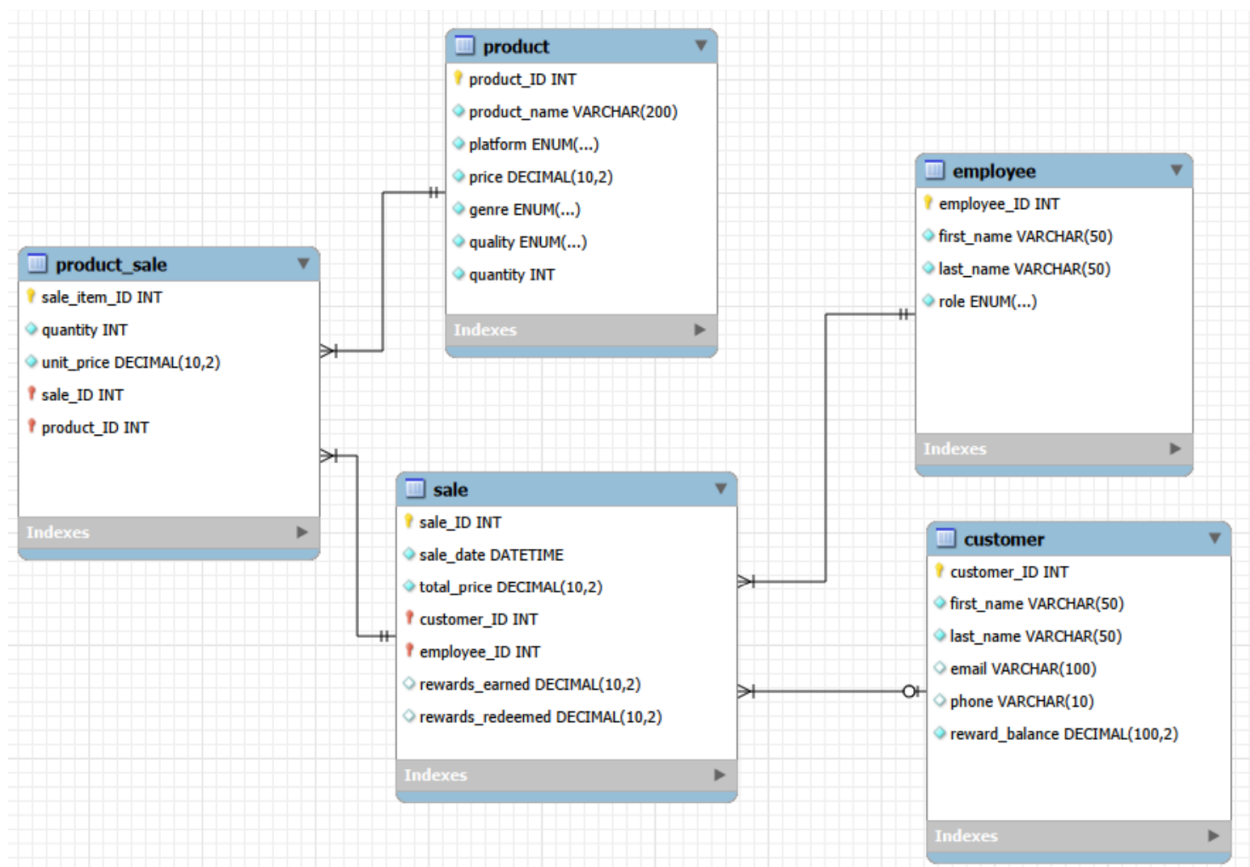
product_sale - Tracks individual products in a sale

- sale_item_ID: int, PK, not null, unique, auto_increment
- sale_ID: int, FK, not null
 - ON DELETE CASCADE
- product_ID: int, FK, not null
 - ON DELETE RESTRICT
- quantity: int, not null
- unit_price: decimal(10,2), not null
- Relationship - Bridges product and sale entities to form M:N relation
 - product_sale to sales (M:1) implementation of sale_ID as FK into product_sale
 - product_sale to product (M:1) implementation of product_ID as FK into product_sale

ERD



Schema



Sample Data

product						
product_ID	product_name	platform	genre	price	quality	quantity
1	The Last of Us Part II	Playstation	Action	30.00	Excellent	9
2	The Last of Us Part II	Playstation	Action	15.00	Poor	18
8	Forza Horizon 5	Xbox	Racing	30.00	Average	18
16	Catan	Board Game	Strategy	35.00	Average	2

customer					
customer_ID	first_name	last_name	email	phone	reward_balance
1	Tom	Hanks	Tomh@yahoo.com		300.00
2	Emma	Watson		3057096482	28.70
3	Robert	Pattinson	RobPat1234@gmail.com	6317694595	9.20
4	Amelia	Earhart	AmeliaFlies@gmail.com	5056441324	58.00
5	Freddie	Mercury	Freddddy@hotmail.com		144.60

employee			
employee_ID	first_name	last_name	role
1	Simone	Biles	Supervisor
2	Johnny	Depp	Stocker
3	Steve	Harvey	Cashier

4	Dolly	Parton	Manager
---	-------	--------	---------

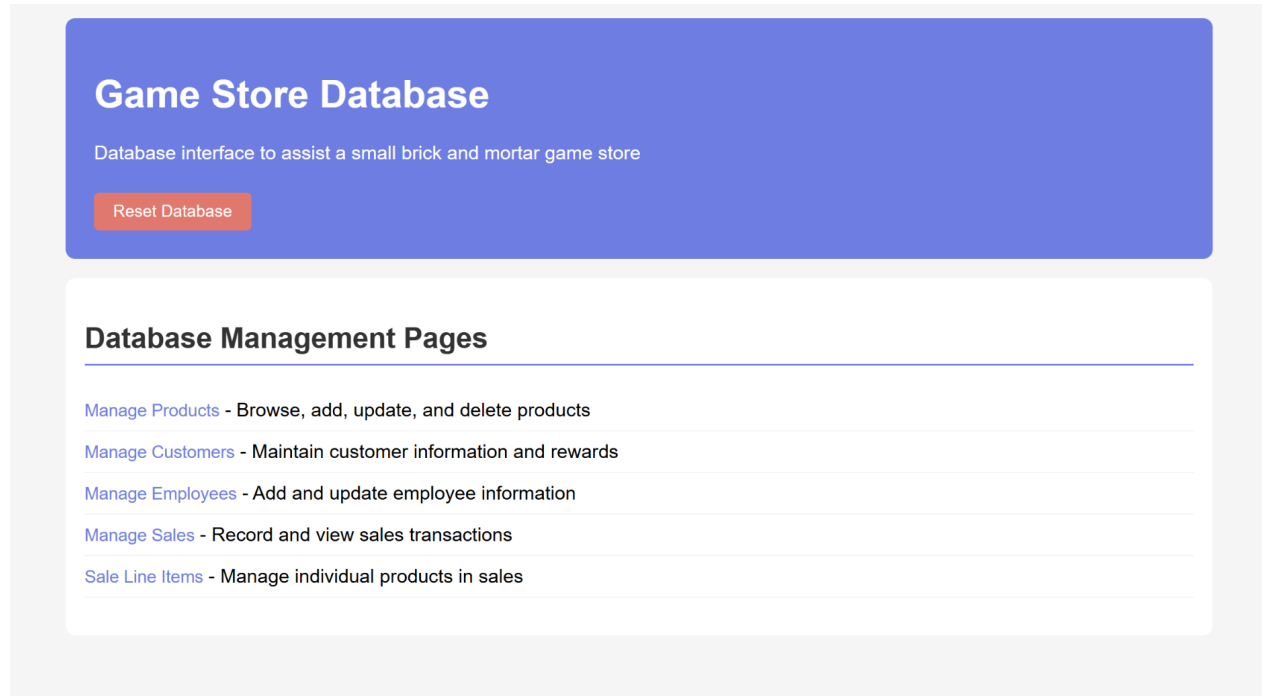
sale						
sale_ID	sale_date	total_price	customer_ID	employee_ID	rewards_earned	rewards_redeemed
1	2026-01-25 11:18	48.00	Null	3	Null	0
2	2026-01-25 12:45	100.00	3	3	10.00	0
3	2026-01-25 22:29	220.00	2	1	22.00	0
4	2026-01-25 14:44	30	Null	3	Null	0
5	2026-01-25 15:51	180	1	1	18.00	162.00

product_sale				
sale_item_ID	sale_ID	product_ID	quantity	unit_price
1	1	1	1	30
2	1	4	1	18
3	2	16	1	35
4	2	18	1	45
5	2	15	1	20

Screen Captures and CRUD operations

Home Page

- No CRUD operations
- Contains Reset button (Also found in the navigation bar on each page)



Products Page

- CREATE/INSERT/ADD NEW Product
- READ/BROWSE/DISPLAY Products
- UPDATE/EDIT Product
- DELETE/REMOVE Product

[Home](#) [Products](#) [Customers](#) [Employees](#) [Sales](#) [Sale Items](#) [Reset Database](#)

Manage Products

Add New Product

Product Name:

Platform:

Genre:

Price:

Quality:

Quantity:

Update Product

Product ID:

Product Name:

Platform:

Genre:

Price:

Quality:

Quantity:

All Products

Product ID	Product Name	Platform	Genre	Price	Quality	Quantity	Actions	
1	The Last of Us Part II	Playstation	Action	\$30.00	Excellent	9	Edit	Delete
2	The Last of Us Part II	Playstation	Action	\$15.00	Poor	18	Edit	Delete
3	The Elder Scrolls V: Skyrim	Playstation	RPG	\$25.00	Near Mint	20	Edit	Delete
4	Minecraft	Playstation	Sandbox	\$20.00	Average	12	Edit	Delete
5	Resident Evil Village	Playstation	Horror	\$18.00	Poor	18	Edit	Delete
6	Tomb Raider: Legacy of Atlantis	Xbox	Adventure	\$23.00	Near Mint	6	Edit	Delete
7	Halo Infinite	Xbox	Shooter	\$40.00	Excellent	11	Edit	Delete
8	Forza Horizon 5	Xbox	Racing	\$30.00	Average	18	Edit	Delete
9	FIFA 23	Xbox	Sports	\$20.00	Average	12	Edit	Delete
10	Mario Kart 8 Deluxe	Nintendo	Racing	\$50.00	Near Mint	2	Edit	Delete
11	Mario Party 9	Nintendo	Party	\$40.00	Excellent	17	Edit	Delete
12	Street Fighter 6	Nintendo	Fighting	\$45.00	Near Mint	12	Edit	Delete
13	The Sims 4	PC	Simulation	\$30.00	Excellent	4	Edit	Delete
14	Portal 2	PC	Puzzle	\$15.00	Near Mint	15	Edit	Delete
15	Subnautica	PC	Survival	\$20.00	Average	7	Edit	Delete
16	Catan	Board Game	Strategy	\$35.00	Average	2	Edit	Delete
17	Catan	Board Game	Strategy	\$0.00	Unsellable	1	Edit	Delete
18	Catan	Board Game	Strategy	\$45.00	Excellent	2	Edit	Delete
19	7 Wonders	Board Game	Strategy	\$40.00	Excellent	19	Edit	Delete
20	Monopoly	Board Game	Party	\$15.00	Poor	15	Edit	Delete
21	UNO	Card Game	Party	\$5.00	Poor	16	Edit	Delete
22	Exploding Kittens	Card Game	Party	\$15.00	Near Mint	7	Edit	Delete
23	Nintendo Switch Pro Controller	Nintendo	Other	\$60.00	Near Mint	11	Edit	Delete
24	Nintendo Switch Pro Controller	Nintendo	Other	\$40.00	Average	15	Edit	Delete
25	Xbox Wireless Controller	Xbox	Other	\$65.00	Near Mint	0	Edit	Delete

Customers Page

- CREATE/INSERT/ADD NEW Customer
- READ/BROWSE/DISPLAY Customers
- UPDATE/EDIT Customer
- DELETE/REMOVE Customer

[Home](#) [Products](#) [Customers](#) [Employees](#) [Sales](#) [Sale Items](#) [Reset Database](#)

Manage Customers

Add New Customer

First Name:

Last Name:

Email:

Phone:

Note: Either email or phone is required

[Add Customer](#)

Update Customer

Customer ID:

First Name:

Last Name:

Email:

Phone:

Note: Either email or phone is required

[Update Customer](#)

All Customers

ID	First Name	Last Name	Email	Phone	Reward Balance	Actions
1	Tom	Hanks	Tomh@yahoo.com	—	\$300.00	Edit Delete
2	Emma	Watson	—	3057096482	\$28.70	Edit Delete
3	Robert	Pattinson	RobPat1234@gmail.com	6317694595	\$9.20	Edit Delete
4	Amelia	Earhart	AmeliaFlies@gmail.com	5056441324	\$58.00	Edit Delete
5	Freddie	Mercury	Fredddy@hotmail.com	—	\$144.60	Edit Delete

Employees Page

- CREATE/INSERT/ADD NEW Employee
- READ/BROWSE/DISPLAY Employees
- UPDATE/EDIT Employee
- DELETE/REMOVE Employee

[Home](#) [Products](#) [Customers](#) [Employees](#) [Sales](#) [Sale Items](#) [Reset Database](#)

Manage Employees

Add New Employee

First Name:

Last Name:

Role: ▼

Add Employee

Update Employee

Employee ID:

First Name:

Last Name:

Role: ▼

Update Employee

All Employees

ID	First Name	Last Name	Role	Actions
1	Simone	Biles	Supervisor	Edit Delete
2	Johnny	Depp	Stocker	Edit Delete
3	Steve	Harvey	Cashier	Edit Delete
4	Dolly	Parton	Manager	Edit Delete

Sales Page

- CREATE/INSERT/ADD NEW Sale
- READ/BROWSE/DISPLAY Sales
- DELETE/REMOVE Sale

[Home](#) [Products](#) [Customers](#) [Employees](#) [Sales](#) [Sale Items](#) [Reset Database](#)

Manage Sales

Record New Sale

Sale Date: 

Employee: 

Customer (Optional): 

Total Price:

Rewards Redeemed:

[Record Sale](#)

Recent Sales

Sale ID	Date	Customer	Employee	Total	Rewards Earned	Rewards Redeemed	Actions
1	1/25/2026, 11:18:00 AM	Walk-in Customer	Steve Harvey	\$48.00	N/A	N/A	Delete
2	1/25/2026, 12:47:00 PM	Robert Pattinson	Steve Harvey	\$100.00	\$10.00	\$0.00	Delete
3	1/25/2026, 10:29:00 PM	Emma Watson	Simone Biles	\$220.00	\$22.00	\$0.00	Delete
4	1/25/2026, 2:44:00 PM	Walk-in Customer	Steve Harvey	\$30.00	N/A	N/A	Delete
5	1/25/2026, 3:51:00 PM	Tom Hanks	Simone Biles	\$180.00	\$18.00	\$162.00	Delete

Sale Items Page

- CREATE/INSERT/ADD NEW Sale Item
- READ/BROWSE/DISPLAY Sale Items
- UPDATE/EDIT Sale Item (Update of a M:N relationship)
- DELETE/REMOVE Sale Item (Delete from a M:N relationship)

[Home](#) [Products](#) [Customers](#) [Employees](#) [Sales](#) [Sale Items](#) [Reset Database](#)

Product Sale Line Items

Add Product to Sale

Sale ID:

Product:

Quantity:

Unit Price:

[Add to Sale](#)

Update Sale Item

Sale Item ID:

Product:

Quantity:

Unit Price:

[Update Sale Item](#)

All Sale Line Items

Item ID	Sale ID	Product ID	Product Name	Platform	Quantity	Unit Price	Line Total	Actions
1	1	1	The Last of Us Part II	Playstation	1	\$30.00	\$30.00	Edit Delete
2	1	4	Minecraft	Playstation	1	\$18.00	\$18.00	Edit Delete
3	2	16	Catan	Board Game	1	\$35.00	\$35.00	Edit Delete
4	2	18	Catan	Board Game	1	\$45.00	\$45.00	Edit Delete
5	2	15	Subnautica	PC	1	\$20.00	\$20.00	Edit Delete
6	3	23	Nintendo Switch Pro Controller	Nintendo	3	\$60.00	\$180.00	Edit Delete
7	3	24	Nintendo Switch Pro Controller	Nintendo	1	\$40.00	\$40.00	Edit Delete
8	4	13	The Sims 4	PC	2	\$30.00	\$60.00	Edit Delete
9	5	10	Mario Kart 8 Deluxe	Nintendo	1	\$50.00	\$50.00	Edit Delete
10	5	11	Mario Party 9	Nintendo	1	\$40.00	\$40.00	Edit Delete
11	5	12	Street Fighter 6	Nintendo	2	\$45.00	\$90.00	Edit Delete